



Design of Everyday Things

CHAPTER 1

*The psychopathology of
everyday things*

What characterizes an object?

What characterizes an object

- Actions
 - `door.push();`
 - `door.pull();`
- State
 - `door.isClosed`
 - `door.isOpen`



*To perform **actions** on an object
and to know its **state**,
We need **clues** that **we** can **relate** to*

Agenda

- Elements of good design
- Fundamental principles of interaction
 - Affordances
 - Signifiers
 - Mapping
 - Feedback
 - Conceptual models

Agenda

- Elements of good design
 - Discoverability
 - Understanding
- Fundamental principles of interaction
 - Affordances
 - Signifiers
 - Mapping
 - Feedback
 - Conceptual models

A good design is ...

Discoverable

- What actions can I perform?
- Where to perform these actions?



A good design is ...

Understandable

- What does it all mean?
- How is the product supposed to be used?
- What do all controls and settings mean?



Discoverability and Understanding

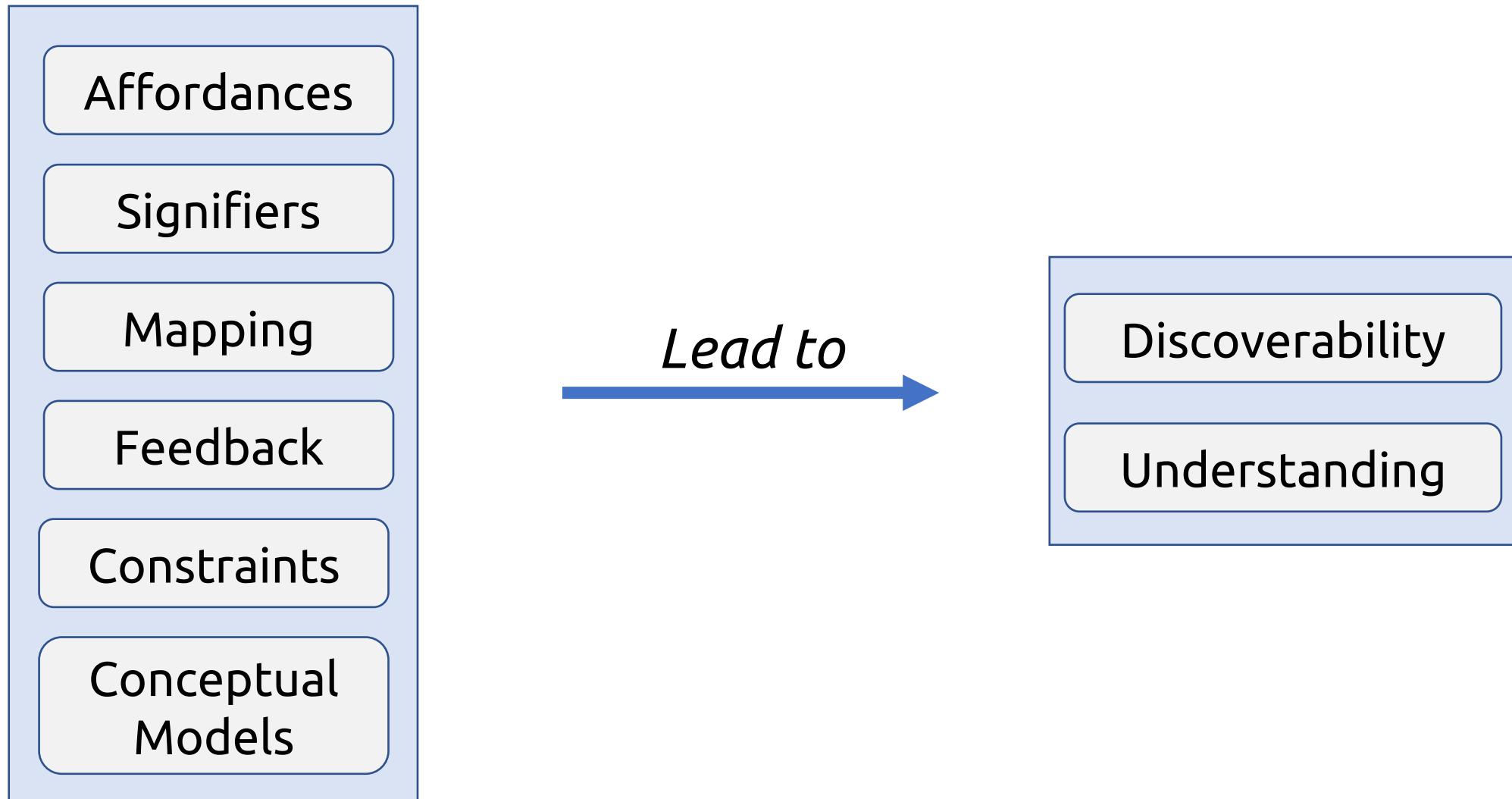
- <https://www.interaction-design.org/literature/topics/signifiers>

How can we design for
discoverability & *understanding*?

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Fundamental principles of interaction



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Affordances

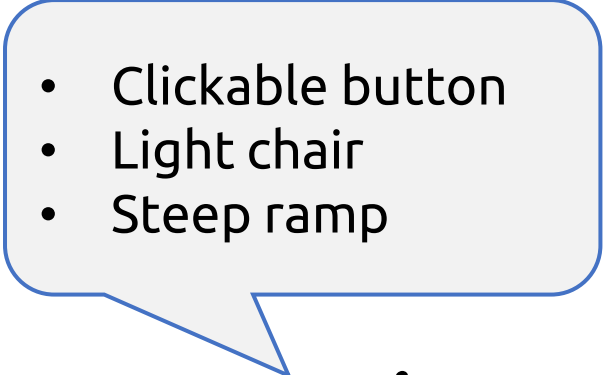
The relationship between the properties of an object and the capabilities of the interacting human with respect to the actions that can be performed on this object by this human

Affordances

Concerned with two sides: the
object and the human

The relationship between the properties of an object and the capabilities of the interacting human with respect to the actions that can be performed on this object by this human

Affordances

- 
- Clickable button
 - Light chair
 - Steep ramp

*The relationship between the **properties of an object** and the capabilities of the interacting human with respect to the actions that can be performed on this object by this human*

Affordances

- Strong
- Tall
- Fully sighted

The relationship between the properties of an object and the capabilities of the interacting human with respect to the actions that can be performed on this object by this human

Affordances

- <Heavy chair, strong human> → The chair affords lifting
- <Touchable screen, no hands> → The touchscreen doesn't afford touching

7 object and the capabilities of the interacting human with respect to the actions that can be performed on this object by this human

Affordances - Example

- What actions can be done with a sheet of glass by an interacting agent, e.g.,
 - Air
 - Light
 - Humans

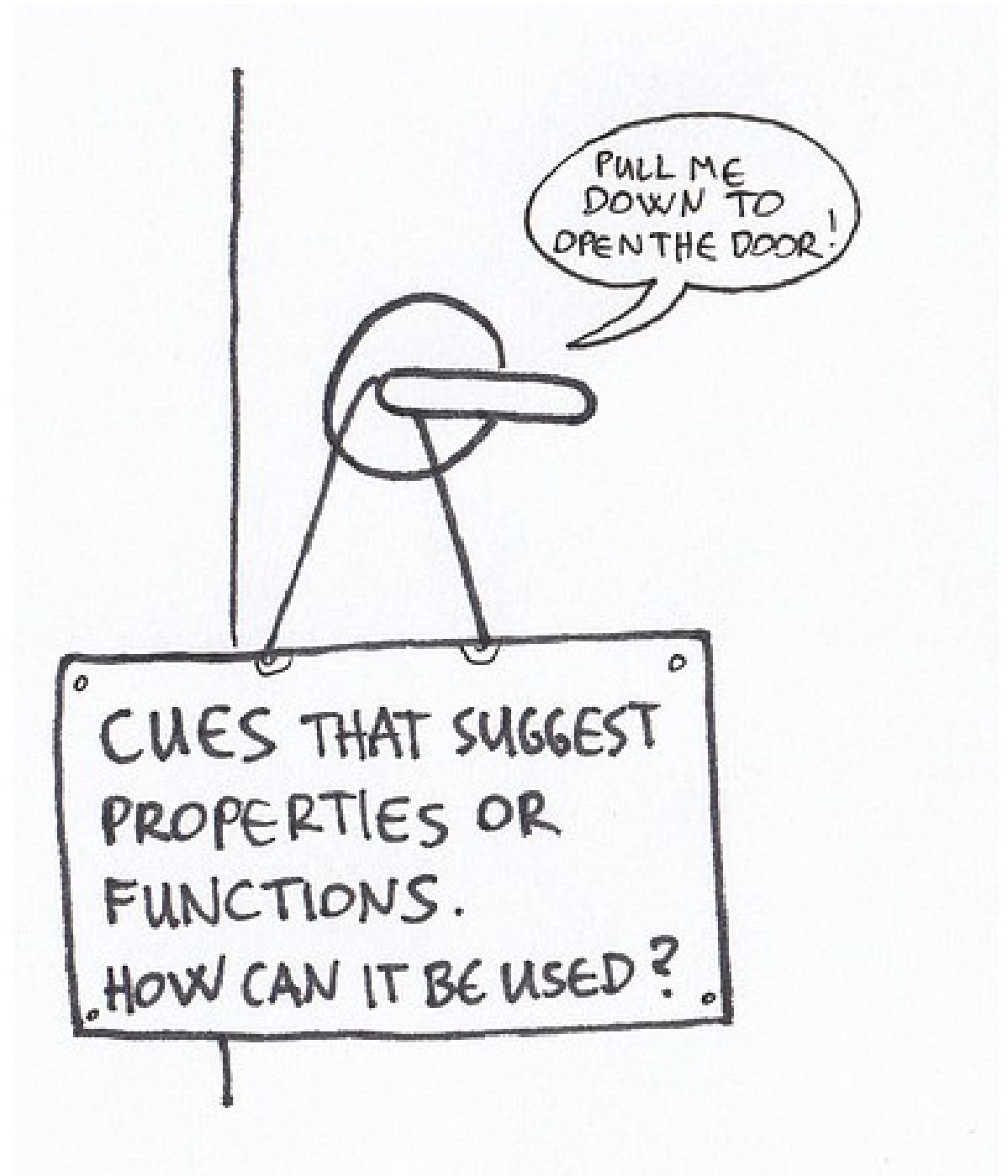


Affordances - Example

- **Glass** affords **transparency**.
- At the same time, its physical structure **blocks the passage of most physical objects**.
 - As a result, **glass affords seeing through** and support, **but not the passage of air or most physical objects** (atomic particles can pass through glass).
- The blockage of passage can be considered an **anti-affordance** — the prevention of interaction.
- To be effective, **affordances and anti-affordances have to be discoverable**—perceivable.
 - If an affordance or anti-affordance cannot be perceived, some means of signaling its presence is required: **Signifier**

Affordances

Cues or signs
communicated by the
objects



Affordances

- If affordances are concerned with actions that are possible given our abilities, then ...

Anti-affordances are concerned with actions that are ***not possible*** given our capabilities

Affordances

Effective affordances are *perceivable*

Affordances

- What makes them perceivable?

Object form × Interaction expectations



Affordances

- What makes them perceivable?

Object form × Interaction expectations



Affordances

- What makes them perceivable?

Object form × Interaction expectations



Affordances

- “In general, when the apparent affordances of an artifact matches its intended use, the artifact is easy to operate. When apparent affordances suggest different actions than those for which the object is designed, errors are common.”

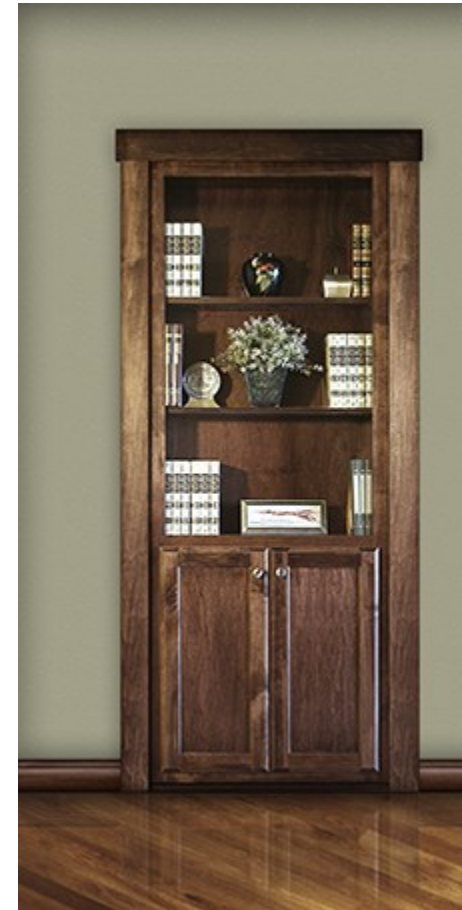
William Gaver (a notable designer)

Hidden Affordances

- In user interfaces without obvious affordances, users often must rely on experience and/or trial and error to determine possible actions.
- For example:
 - Users hover/click on suspected drop-down menus.
 - Hidden menus in apps
 - Keyboard shortcuts
 - Secret Compartment in Furniture

Hidden Affordances

- When hidden affordances are put to use
- E.g., Murphy Door
 - A common type of Murphy hidden door is a bookshelf door, where the door looks like a bookshelf.
 - When the right mechanism is used, the entire bookshelf swings open to reveal a hidden room or passage.



Hidden Affordances

- When hidden affordances are put to use



Hidden Affordances

- When hidden affordances are put to use
 - Game design
 - Ads

False affordances

An object's characteristics suggest users can do something they can't. For example:

- Drawers That Don't Open
- Underlined text that isn't a link
- Non-clickable Buttons on Websites



False affordances



Affordances

<https://www.interaction-design.org/literature/topics/affordances>

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Signifiers

- Object: a door
- Agent: a human with functioning hands
- What's the affordance?
- Where can I do the action?



Signifiers – The *clues*

“Any mark or sound, any perceivable indicator that communicates appropriate behavior to a person”

Signifiers – The *clues*

“Any mark or sound, any perceivable indicator that communicates appropriate behavior to a person”

Signifiers – The *clues*

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Signifiers – The *clues*

“Any mark or sound, any perceivable indicator that communicates appropriate behavior to a person”

The communication medium has to be ***compatible*** & ***perceivable*** to ***that person***

Signifiers – The *clues*

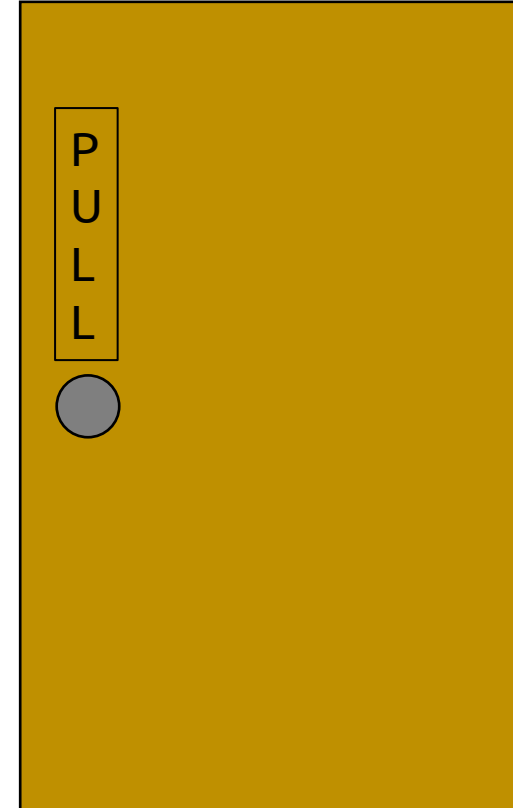
- Your job as a designer is to ...

Communicate meaningful clues to your user group about the use of the design to highlight the design purpose

Signifiers – The *clues*



Perceived



Explicit

Problem Doors: Signifiers Are Needed

A.



C.



B.



Problem Doors: Signifiers Are Needed

- Door hardware can signal whether to push or pull without signs, but the hardware of the two doors in the upper photo, A, are identical even though one should be pushed, the other pulled.
 - The flat, ribbed horizontal bar has the obvious perceived affordance of pushing, but as the signs indicate, the door on the left is to be pulled, the one on the right is to be pushed.
- In the bottom pair of photos, B and C, there are no visible signifiers or affordances.
 - How does one know which side to push? Trial and error.
- When external signifiers—signs— have to be added to something as simple as a door, it indicates bad design.

Sliding Doors: Seldom Done Well

A.



B.



C.



Sliding Doors: Seldom Done Well

- Sliding doors are seldom signified properly.
- The top two photographs show the sliding door to the toilet on an Amtrak train in the United States. The handle clearly signifies “pull,” but in fact, it needs to be rotated and the door slid to the right.
- The owner of the store in Shanghai, China, Photo C, solved the problem with a sign. “DON’T PUSH!” it says, in both English and Chinese.
- Amtrak’s toilet door could have used a similar kind of sign.

Affordances vs. Signifiers

Signifiers *expose* affordances,
they *save* people from trial and error

Examples of signifiers in UI design

- Button states

- Button states provide visual cues to users about the functionality of the button and whether or not it can be interacted with.

- Icons

- a magnifying glass icon is commonly used to indicate a search function,
- a trash can icon is used to indicate a delete function.

- Color

- a red color may be used to indicate an error or warning,
- a green color may be used to indicate success.

- Feedback

- Feedback, such as animations, sounds, or haptic feedback, can also act as signifiers to indicate successful completion of an action or an error.

Affordances & Signifiers in user interfaces



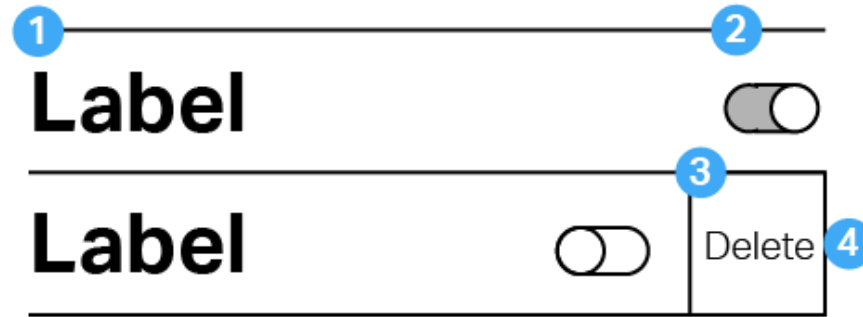
Three signifiers reveal this object's enhanced affordance

1   Remove/Add (New)

2 Label

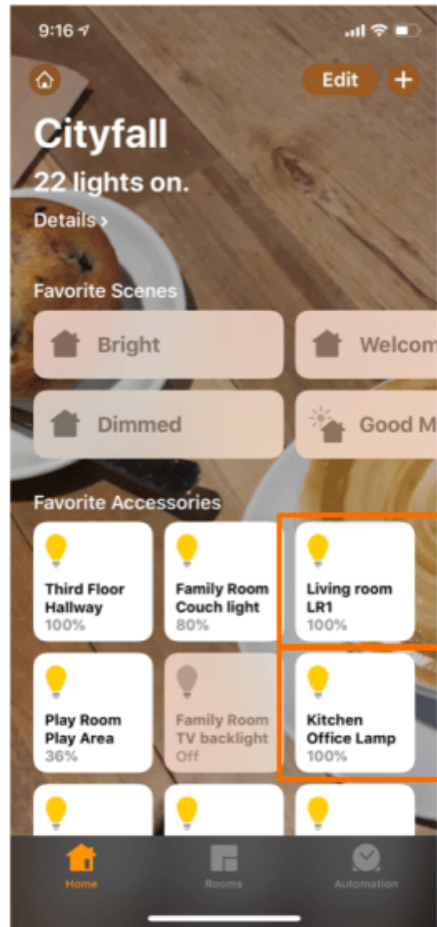
3  Draggable

Affordances & Signifiers in user interfaces



- 1 Label indicates function/content
- 2  Signifies On/Off
- 3 Discoverable affordance through swipe
- 4 Label is instructional

Affordances & Signifiers in user interfaces



Affordance

- The button itself (invisible tap target that takes up set space on touch system (the primary affordance), not the visual representation of a button)

Signifiers

- The visual representation of a button
- The lightbulb icon to tell you it's a lightbulb you are controlling
- The label to tell you which room the light is in
- The percentage to tell you how bright the bulb is

Signifiers on a Touch Screen

- The arrows and icons are signifiers:
 - they provide signals about the permissible operations for this restaurant guide.
 - Swiping left or right brings up new restaurant recommendations.
 - Swiping up reveals the menu for the restaurant being displayed;
 - swiping down, friends who recommend the restaurant.



Difference between Affordances, Perceived Affordances and signifiers

- <https://www.interaction-design.org/literature/topics/signifiers>

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Mapping



Mapping



Mapping

The spatial relationship between the control and what's being controlled

Mapping

The spatial relationship between the control and what's being controlled

Mapping

The spatial relationship between the control and what's being controlled

Mapping

*The spatial relationship between the control and
what's being controlled*

Mapping & conceptual models

- **Steering a car**
 - The top of the steering wheel moves to the same direction as the car's
- **Moving a computer mouse**
 - X-axis of the mouse = X-axis of the screen
 - Z-axis of the mouse = Y-axis of the screen
- **Dimming lights**
 - Up = more light
 - Down = less light

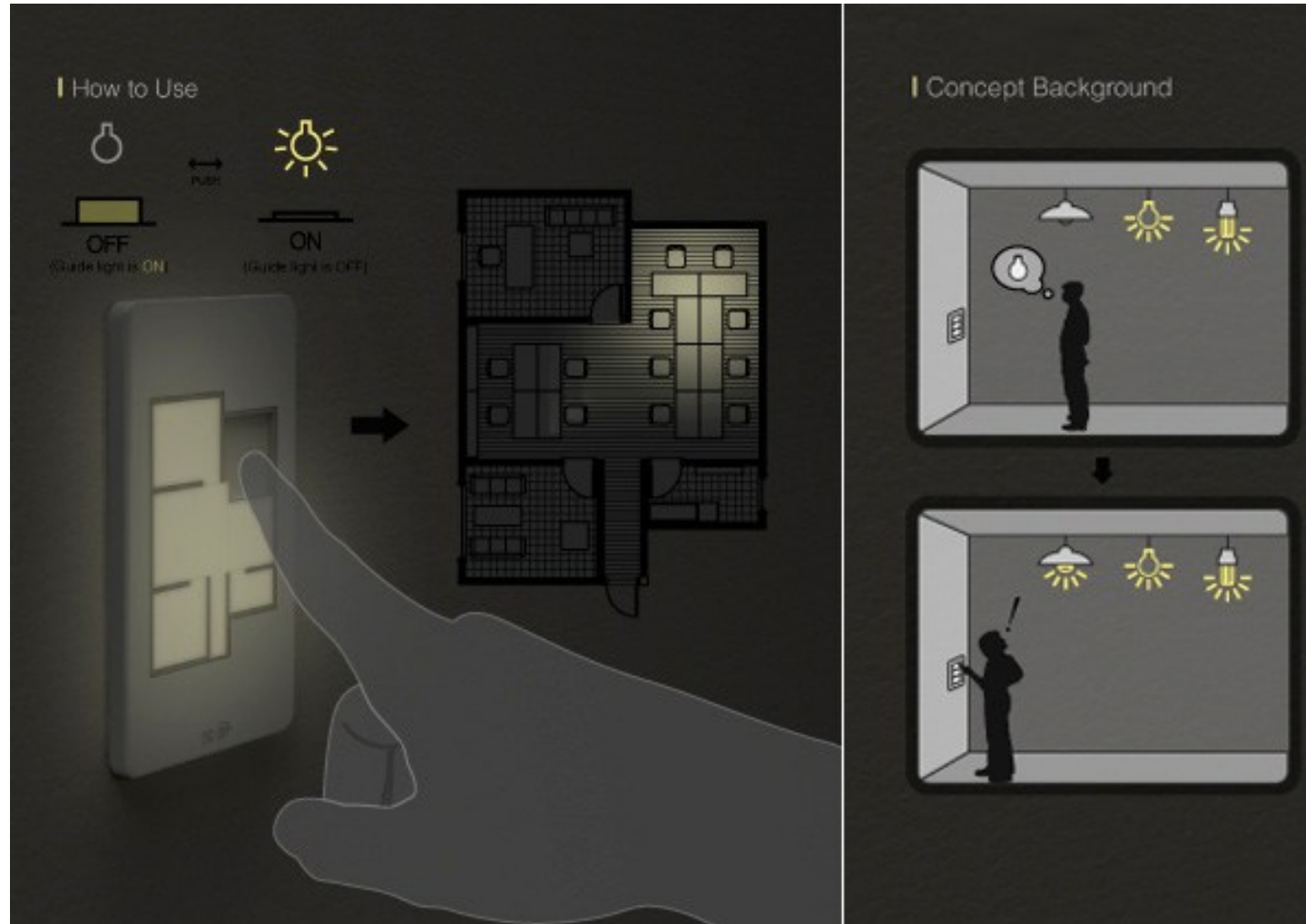
Mapping & conceptual models

Clear and ***understandable*** conceptual models
help us ***remember*** the mapping relationship

Natural mapping



Natural mapping: Floor Plan Light Switches



Good Mapping: Automobile Seat Adjustment Control

- This is an excellent example of natural mapping.
- The control is in the shape of the seat itself: the mapping is straightforward.
 - To move the front edge of the seat higher, lift up on the front part of the button.
 - To make the seat back recline, move the button back.
- This particular control is from Mercedes-Benz, but this form of mapping is now used by many automobile companies.
- The same principle could be applied to much more common objects.

Natural mapping: Automobile Seat Adjustment Control



Mapping & Cultural views

What is natural for one culture is not necessarily natural for another

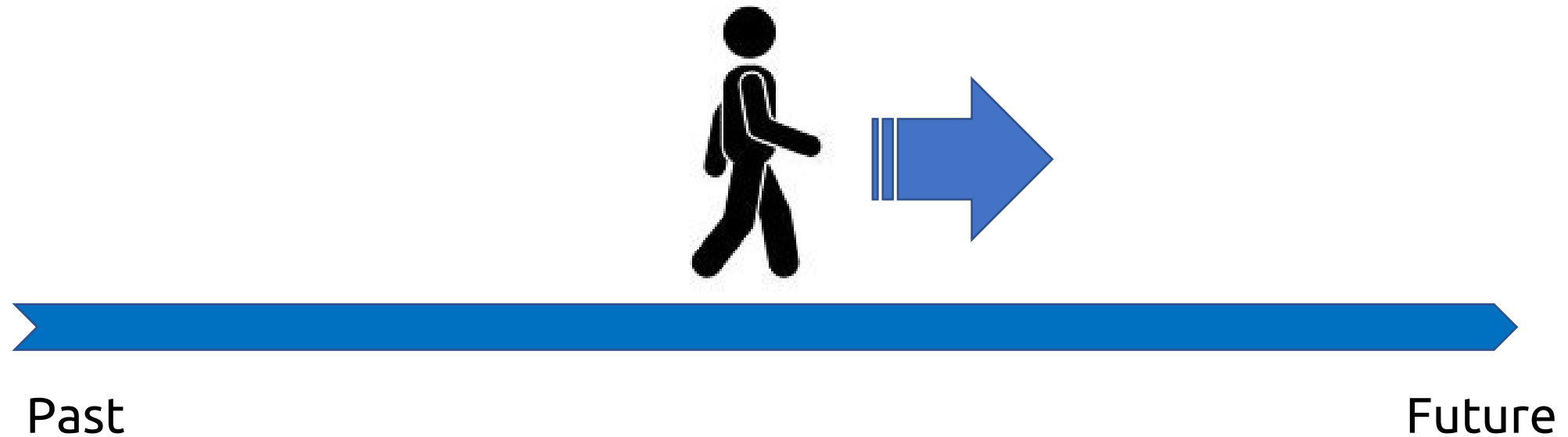
Mapping & Cultural views

- To get to the next slide where should I scroll?



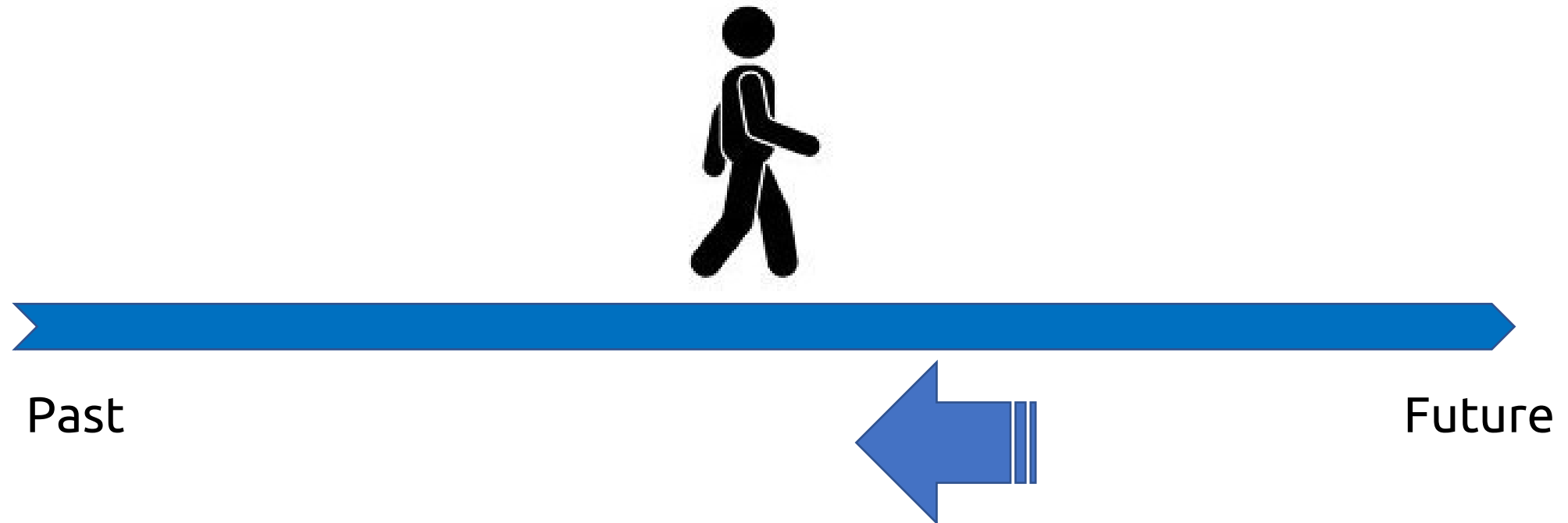
Mapping & Cultural views

- Culture A: we move and timeline is fixed → scroll *up* for next



Mapping & Cultural views

- Culture B: Timeline moves and we are fixed → scroll *down* for next



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Feedback



Feedback



Why do we do that?

Feedback



Why do we feel this way?

Feedback

“Communicating the results of an action”

Feedback can be ...



Haptic



Visual



Auditory

Good feedback is *perceivable*

- Uses the best channel given the current situation and the person's abilities
- Beeping sound feedback for deaf people?
- Visual Google Maps directions while driving?

Good feedback is *immediate*



0.1 seconds

Instantaneous response



1 second

Seamless



$1 < t \leq 10$ seconds

Keeps their attention



> 10

Difficult to remember
what happened

Good feedback is *informative*

- What do all these meaningless beeps and flashes mean?!



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My coffee maker just keeps beeping? Whats that mean? wil not brew or anything.

Posted by [jwvsimmons](#) on Dec 13, 2010

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2,332

people viewed this question

Ask a Question

Usually answered in minutes!

What do you need help with?

Good feedback is *thoughtful*

- Selects the right channel at the right time

Google Maps
+
Making a call
=
Annoying!



Good feedback is *moderate*

- Too much feedback is distracting or even overwhelming

Good feedback is *unobtrusive*

- The main goal is *acting on the problem* and *not attending* to the feedback

Association for Computing Machinery (ACM) ?

[-] 'Organisation' is a required field.

1. A post office address cannot be accepted as an organisation address.

UNITED STATES ▾

2 Penn Plaza, ?

[-] 'Address' is a required field.

Suite 701

▾

ss contact.

[-] 'Family name' is a required field.

Feedback

- All actions have to be confirmed
- Must be immediate
- Must be informative
- Preferably non-distracting and unobtrusive

Feedback



LOADING...

Agenda

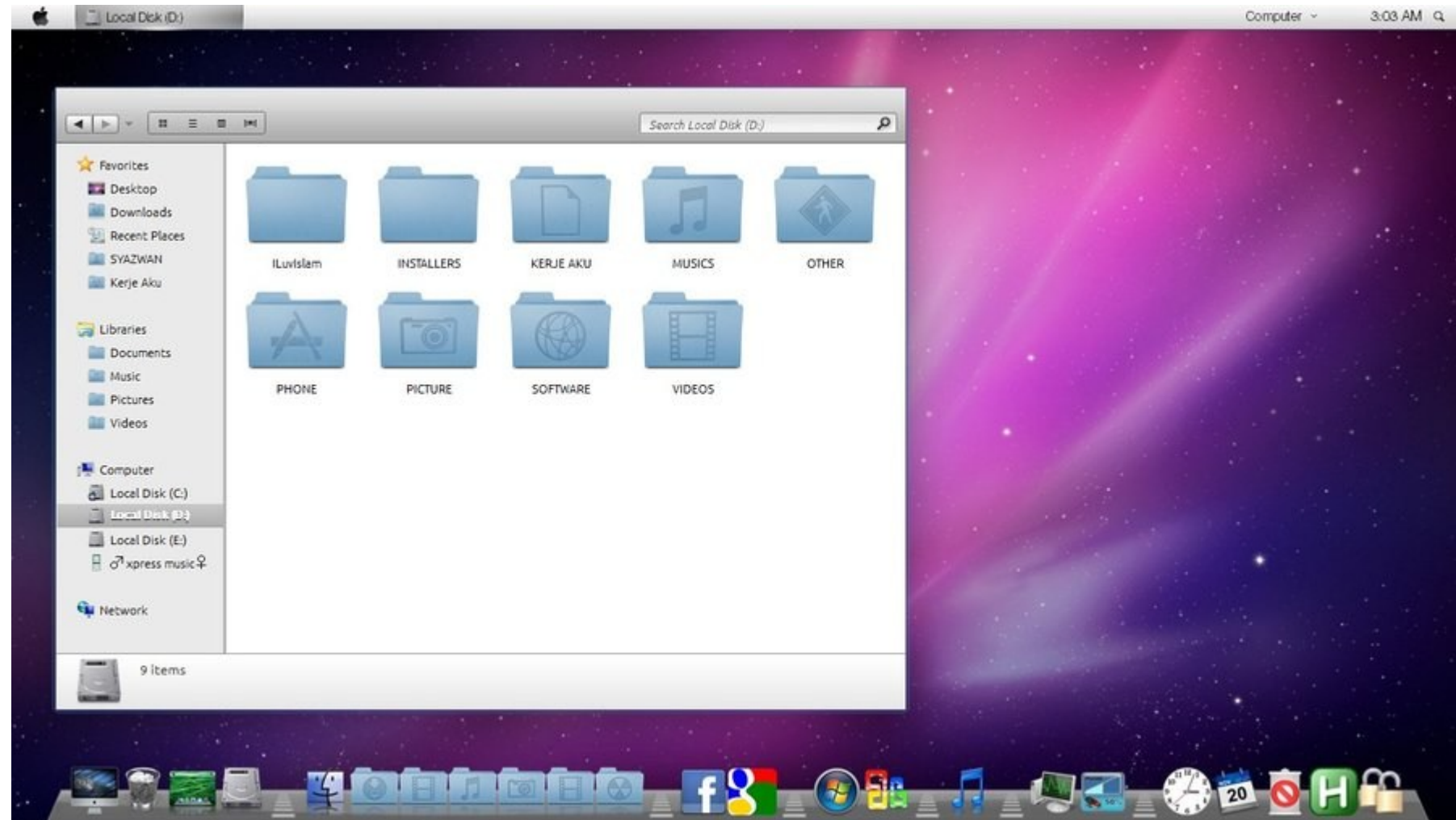
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Conceptual models

“An explanation, usually simplified, of how something works”

Conceptual models

Simplified,
inaccurate,
and
incomplete,
but useful



Conceptual models are built from the *system image*

The physical structure

Affordances, signifiers, and mapping

Interaction with the device

Feedback

Prior knowledge

- Past experiences
- Online forums
- ...

Conceptual models are built from the
system image

*It's up to the system image to convey the designer's
intentions*